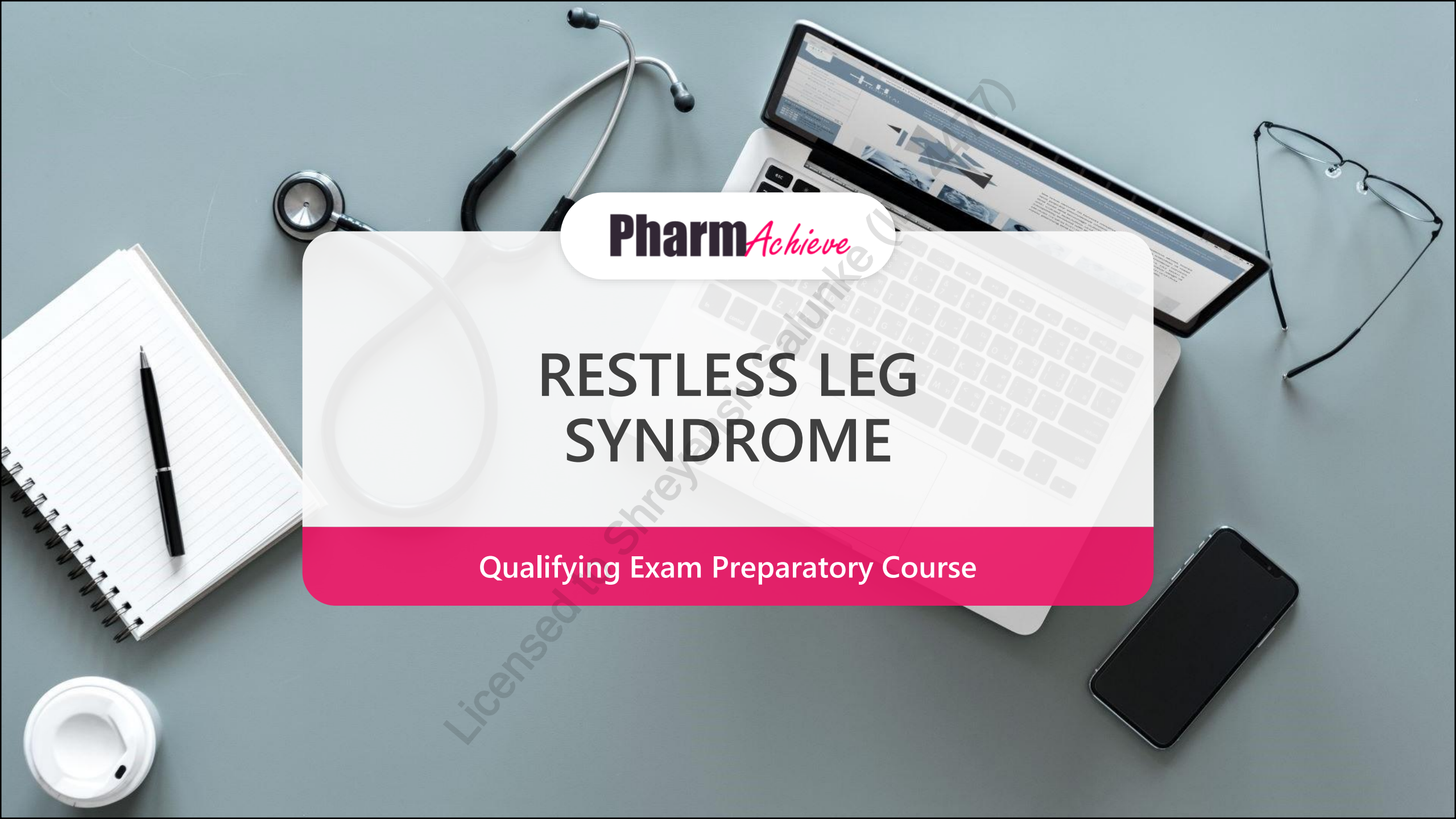




Pharm*Achieve*

RESTLESS LEG SYNDROME

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

ADHD Attention Deficit Hyperactivity Disorder

BUN Blood Urea Nitrogen

CBC Complete Blood Count

PLMS Periodic Limb Movements of Sleep

RLS Restless Legs Syndrome

PATHOPHYSIOLOGY

- Restless Legs Syndrome (RLS) is also known as **Willis-Ekbom disease**
- A **neurologic condition** in which patients experience **irresistible urges to move their legs along with unpleasant sensations such as tugging or pulling**
- Patients usually experience symptoms in the **evening**, which causes **sleep disturbances** and reduces overall quality of life
- **Primary RLS:** hereditary or idiopathic
- **Secondary RLS:** stems from another underlying medical condition (i.e. diabetes, end-stage renal disease, iron deficiency, MS, Parkinson disease) or is linked to the use of medications

CLINICAL PRESENTATION

Urge to move legs (or arms and trunk in severe cases) along with uncomfortable sensations in those areas

- Patients describe the unpleasant sensations as “burning, nagging, aching, electric-current, creepy-crawly, itching-bones”
- Commonly bilateral (can be unilateral on occasion)
- Worsens during inactivity or at rest and improves with activity
- Worsens or only occurs during the evening/night

RISK FACTORS AND CAUSES

Risk factors

- Family history
- Gender (women > men)
- Increasing age
- Pregnancy (symptoms usually disappear after giving birth)
- Peripheral neuropathy (damage to nerves of hands and feet)
- Attention deficit hyperactivity disorder (ADHD)

Medically-induced causes (secondary RLS)

- Diabetes mellitus
- End-stage renal disease
- Iron deficiency
- Multiple sclerosis
- Parkinson's disease
- Venous insufficiency

Drugs that can worsen symptoms (secondary RLS)

- Alcohol, caffeine, nicotine
- Anti-depressants (except Bupropion)
- Antipsychotics (commonly quetiapine, olanzapine, clozapine)
- Antiemetics (e.g. metoclopramide)
- Antiepileptics (commonly topiramate)
- 1st generation anti-histamines

DIAGNOSIS

FYI

- There are no specific tests to diagnose RLS
- Examinations include sleeping patterns and PLMS, lab tests (e.g. CBC, electrolytes, BUN, creatinine, fasting glucose, serum iron, ferritin, iron saturation)

All 5 of the following criteria must be met to diagnose RLS:

1. Urge to move the legs, usually accompanied or caused by unpleasant sensations in the legs
2. Symptoms begin or worsen during periods of rest/inactivity
3. Relief of symptoms (partial or complete) with movement
4. Symptoms worsen/may only occur in the evening or at night
5. Symptoms not solely accounted for by another medical or behavioural condition

CLASSIFICATIONS OF RLS

Intermittent RLS

Symptoms occur < 2 times/week but disrupt the patient's daily living, requiring on-demand therapy

Chronic persistent RLS

Symptoms occur ≥ 2 times/week with moderate-severe distress, requiring daily therapy

Refractory RLS

Unresponsive to maximum tolerated doses of first-line therapy or inadequate response due to augmentation

GOALS OF THERAPY

- ✓ Improve motor restlessness
- ✓ Improve sleep
- ✓ Improve daily functioning of patients
- ✓ Reduce discomfort
- ✓ Reduce PLMS
- ✓ Reduce risk for rebound (recurrence of symptoms during the night or early morning coinciding with dose wearing off) or augmentation (worsening symptoms) with drug therapy

NON-PHARMACOLOGICAL MEASURES

- Mental alertness activities during times of boredom to minimize symptoms (e.g. crossword puzzles)
- Avoid alcohol, caffeine and nicotine
- Stretch, exercise, yoga
- Massage
- Hot baths
- Minimize aggravating factors: sleep deprivation, medications

Non-drug therapy can be used in milder cases or as an adjunct in moderate-severe cases.
Drug therapy is usually needed in moderate-severe cases.

PHARMACOLOGICAL MEASURES

- Evidence that 25-30% of RLS patients are **iron deficient**
- Patients should be assessed for iron deficiencies and a trial of oral iron therapy should be considered in those with a deficiency (ferritin levels ≤ 75 mcg/L in adults and < 50 mcg/L in children)

Overview of drug therapy for RLS

Intermittent RLS

- Levodopa preparations
- Benzodiazepines
- Low-potency opioids

Chronic Persistent RLS

- GABA derivatives
- Non-ergot dopamine agonists

Refractory RLS

- Combination therapy with drugs from different classes
- Opioid monotherapy

INTERMITTENT RLS

May be sufficiently managed by non-pharmacological measures. However, if symptoms persist, the following may be considered:

Levodopa Preparations

- Preferred for symptoms occurring at night
- Limitation: **risk of rebound** and **augmentation** over time → therefore, should not be used as standard therapy

Benzodiazepines

- Useful for improving quality of sleep
- Limitation: dependence potential and risk of falls in elderly
- Temazepam: short/immediate duration of action → recommended for the elderly
- Clonazepam: most studied, however has many adverse effects → use is not recommended as per newer guidance

Opioids

- Effective in those whose symptoms are not relieved by other medications
- Limitation: dependence potential → preferred use of low-potency opioids e.g. codeine or tramadol

CHRONIC PERSISTENT RLS

GABA derivatives (e.g. gabapentin, pregabalin)

- Preferred agent
- Useful in patients with comorbid insomnia, anxiety or pain
- Pregabalin should be considered in those who develop augmentation with a dopamine agonist
- Side effects: Drowsiness, unsteady gait (especially in elderly)

Dopamine agonists (e.g. pramipexole, ropinirole and rotigotine)

- Alternative if GABA agents ineffective or not tolerated
- May be considered in patients with comorbid depression, excessive weight, cognitive impairment, at risk for pulmonary depression or falling
- Improves sleep quality and reduces amount of periodic limb movements during sleep
- Adverse effects: nausea, sedation, lightheadedness
- Limitations to use: **high risk of compulsive behaviors** (6-17%) and **augmentation**

SPECIAL POPULATIONS

Pregnancy and Lactation

- Symptoms of RLS may occur or worsen, with severity peaking during the third trimester
- Symptoms resolve after birth
- Non-pharmacological management preferred
- If required, pharmacological agents are reserved in patients with severe RLS after the first trimester
- **Drug therapy during pregnancy:** benzodiazepines (i.e. clonazepam), levodopa/carbidopa (**avoid** levodopa/benserazide) preparations, oxycodone in severe refractory RLS
- **Drug therapy during lactation:** benzodiazepines (i.e. clonazepam), gabapentin, tramadol in severe refractory RLS
- Levodopa preparations and dopamine agonists should be **avoided** during lactation

Children

- Preferred treatment is iron replacement (e.g. oral ferrous sulfate once daily before breakfast)
- Drug therapy can be considered if iron replacement not sufficient: GABA derivatives are first-line with benzodiazepines following
- Dopamine agonists can be considered, however preferably avoided

MONITORING PARAMETERS

Parameter	Degree of Change	Timeframe
Symptoms of RLS (e.g. urge to move the legs, unpleasant sensations)	Reduce/Improve	Ongoing
Patient comfort and sleeping	Improve	Ongoing



Generally, reassessment should occur every 3 months after initiation and annually for stable patients

TAKEAWAY

- RLS is also known as Willis-Ekbom disease and presents as an **irresistible urge to move the legs**
- **Three subtypes of RLS:** intermittent, chronic persistent and refractory
- Associated with: diabetes, ESRD, iron deficiency, multiple sclerosis, Parkinson's, pregnancy, venous insufficiency

Non-pharmacological options

- Caffeine, alcohol, nicotine avoidance
- Mental alertness activities
- Stop medications that may contribute to symptoms (e.g. antidepressants, antipsychotics, dopamine-blocking antiemetics, first gen antihistamines)

Mainstay drug therapy includes

- Oral iron replacement if ferritin levels are ≤ 75 mcg/L
- Intermittent RLS: on-demand therapy of levodopa, opioids or benzodiazepines
- Chronic persistent RLS: GABA derivatives preferred
- Refractory RLS: combination therapy or opioid monotherapy

- **If one drug is ineffective or not tolerated, switch to another agent in the same class**
- **If ≥ 1 drug(s) in the same class are ineffective or not tolerated, switch to another class**

CASE 1

MX is a 35-year-old male who weighs 180lbs and was diagnosed with intermittent RLS 5 years ago. His past medical history is significant for anxiety. He currently takes Carbidopa/Levodopa 25mg/250mg at night PRN, bupropion XL 150mg PO OD, escitalopram 10mg PO OD and Benadryl® (diphenhydramine) 50mg ON to help him sleep. MX smokes 8-12 cigarettes a day, doesn't drink alcohol and on average drinks 4 cups of coffee a day. He works as a software developer for a large firm and remains seated for most of the day. He tries to take breaks and walks around every couple of hours as recommended by his pharmacist.

1. Which of the following can worsen MX's RLS?
 - a) Bupropion
 - b) Benadryl® (diphenhydramine)
 - c) Alcohol
 - d) Taking walking breaks at work



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2. Which of the following is correct when counselling a patient starting on Carbidopa/Levodopa?
- a) Carbidopa/Levodopa is best taken on an empty stomach
 - b) Carbidopa/Levodopa is best taken with fatty foods
 - c) Carbidopa/Levodopa is best taken with a high protein meal
 - d) Food does not affect Carbidopa/Levodopa



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3. MX has noticed his symptoms have become more persistent and severe, occurring more than 4 times/week. Which of the following is the best next course of action?
- a) Continue Carbidopa/Levodopa
 - b) Discontinue Carbidopa/Levodopa and initiate gabapentin
 - c) Discontinue Carbidopa/Levodopa and initiate clonazepam
 - d) Discontinue Carbidopa/Levodopa and initiate ropinirole



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4. What can MX do to improve his symptoms during work?
- a) Take walking breaks
 - b) Remain mentally alert by doing a puzzle
 - c) Eliminating his intake of coffee
 - d) Changing to a more ergonomically made chair



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CHANGE LOG

May 2025

- Slide 14: Updated pregnancy and lactation information based on UpToDate and guideline
- Slide 21: Add reference

July 2025

- Minor edits on slides 13 & 14

November 2025

- No content changes made

April 2026

- Slide 12: Updated recommendations
- Slide 13: Updates to dopamine agonists place in therapy
- Slide 16: Minor update to takeaway
- Slide 21: Updated references